# Ⅲ. Tangent vectors

## 1. Tangent vectors on manifold

**Def Ⅲ.1.1** $\mathbb{R}^n_a = \{a\} \times \mathbb{R}^n \quad a \in \mathbb{R}^n$ is called the geometric tangent space to $\mathbb{R}^n$ at $a$.

A map $w \quad C^\infty(\mathbb{R}^n) \to \mathbb{R}$ is called a derivation at $a$.

Iff it is linear over $\mathbb{R}$ and $w(fg) = f(a)wg + g(a)wf$.

Let $T_a\mathbb{R}^n$ denote the set of all derivations of $C^\infty(\mathbb{R}^n)$ at $a$.

Then $T_a\mathbb{R}^n$ is the vector space under the operations $(w_1+w_2)f = w_1 f + w_2 f$

$(cw)f = c(wf)$

**Prop. Ⅲ.1.2.** $a \in \mathbb{R}^n$. (a). $w \in T_a\mathbb{R}^n \quad f \in C^\infty(\mathbb{R}^n)$. If $f$ is constant, then $wf = 0$

(b) If $f(a) = g(a) = 0$. then $w(fg) = 0$

(c) $\forall v_a \in \mathbb{R}^n_a$. the map $D_v|_a : C^\infty(\mathbb{R}^n) \to \mathbb{R} \qquad f \mapsto \left.\frac{d}{dt}\right|_{t=0} f(a+tv)$ is a derivation

(d) The map $v_a \mapsto D_v|_a$ is an isomorphism from $\mathbb{R}^n_a$ to $T_a\mathbb{R}^n$.

proof: (a), (b) is trivial.

(c) $D_v|_a(fg) = D_v|_a(f)\cdot g + D_v|_a(g) f$.

(d) $\varphi: v_a \mapsto D_v|_a$. 1°. $\varphi(v_1+v_2) = D_{v_1+v_2}|_a = D_{v_1}|_a + D_{v_2}|_a$. Thus it's linear.

2°. Injective. write $v_a = v^i e_i|_a$, $0 = D_v|_a(x^j) = v^i \frac{\partial}{\partial x_i}(x^j)\Big|_{x=a} = v^j$. (let $f: \mathbb{R}^n \to \mathbb{R}$, $x \mapsto x^j$)

$\Rightarrow v_a = 0$ Thus $\varphi$ is injective.

3°. Surjective. $\forall w \in T_a\mathbb{R}^n$. let $v^i = w(x^i)$. we show that $w = D_v|_a$.

By Taylor's thm, $f(x) = f(a) + \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a)(x^i - a^i) + \sum_{i,j=1}^n (x^i - a^i)(x^j - a^j)\int_0^1 (1-t)\frac{\partial^2 f}{\partial x^i \partial x^j}(a + t(x-a))$

$\Rightarrow wf = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a)\, w(x^i) = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a)\cdot v^i = D_v|_a f$.

Thus $w = D_v|_a$ □

**Coro. Ⅲ.1.3.** $\forall a \in \mathbb{R}^n$. $\left.\frac{\partial}{\partial x^1}\right|_a, \ldots, \left.\frac{\partial}{\partial x^n}\right|_a$ defined by $\left.\frac{\partial}{\partial x^i}\right|_a f := \frac{\partial f}{\partial x_i}(a)$

forms a basis for $T_a\mathbb{R}^n$.

proof: $\left.\frac{\partial}{\partial x^i}\right|_a = De_i|_a$. □.

Def III.1.4. $M^n$ be a smooth manifold. $p \in M^n$

A linear map $v : C^\infty(M^n) \to \mathbb{R}$ is called a derivation at $p$ if $v(fg) = v(f)\,g(p) + v(g)\,f(p)$ $\forall f, g \in C^\infty(M^n)$.

Def III.1.5 All derivations at $p$ form a set, call $T_pM$, is the tangent space to $M$ at $p$. $\forall v \in T_pM$ is called the tangent vector at $p$

Def III 1.6 If $M$ & $N$ are smooth manifolds. $F: M \to N$ be a smooth map.

$\forall p \in M$. define a map $dF_p : T_pM \to T_{F(p)}N$

$$v \mapsto dF_p(v).$$

s.t. $\forall f \in C^\infty(N)$ $dF_p(v)(f) = v(f \circ F)$

Then $dF_p$ is called the differential of $F$ at $p$

Rmk. From the definition, we immediately have. $\forall f, g \in C^\infty(N)\cdot$, $v \in T_pM$

$$dF_p(v)(fg) = v\big((f\circ F)(g\circ F)\big)$$
$$= v(f\circ F)(g\circ F(p)) + v(g\circ F)(f\circ F(p))$$
$$= dF_p(v)(f)\,g(F(p)) + dF_p(v)(g)\,f(F(p))$$

Prop. III.1.7. Let $M, N$ & $P$ be smooth manifolds (with boundary). Let $F: M \to N$, $G: N \to P$ be smooth maps

let $p \in M$. Show that:

(1) $dF_p : T_pM \to T_{F(p)}N$ is linear

(2) $d(G\circ F)_p = dG_{F(p)} \circ dF_p : T_pM \to T_{G\circ F(p)}P$.

(3). $d(Id_M)_p = Id_{T_pM} : T_pM \to T_pM$

(4) If $F$ is a ~~differen~~ diffeomorphism, then $dF_p : T_pM \to T_{F(p)}N$ is an isomorphism. And $(dF_p)^{-1} = d(F^{-1})_{F(p)}$.

pf: (1) $\forall v_1, v_2 \in T_pM$., $dF_p(v_1+v_2)(f) = (v_1+v_2)(f\circ F) = v_1(f\circ F) + v_2(f\circ F)$
$\{ f \in C^\infty(N)$ $= dF_p(v_1)(f) + dF_p(v_2)(f)$.

$dF_p(kv)(f) = k\,v(f\circ F) = k\,dF_p(v)(f)$

$\Rightarrow dF_p$ is linear.

(2) $d(G\circ F)_p(v)(f) = v(f\circ G\circ F)$ $\forall f \in C^\infty(p)$., $v \in T_{G\circ F(p)}P$

and $dG_{F(p)} \circ dF_p(v)(f) = dF_p(v)(f\circ G) = v(f\circ G\circ F)$.

$\Rightarrow d(G\circ F)_p = dG_{F(p)} \circ dF_p$

(3) $d(Id_M)_p(v)(f) = v(f\circ Id_M) = v(f) = Id_{T_pM}(v)(f)$ $\forall f \in C^\infty(M)$, $v \in T_pM$

$\Rightarrow d(Id_M)_p = Id_{T_pM}$

(4). Injective: Suppose that $dF_p(v_1) = dF_p(v_2) \Rightarrow \forall f \in C^\infty(N)$, $v_1(f\circ F) = v_2(f\circ F)$

Since $F$ is a diffeomorphism, $\forall g \in C^\infty(M)$ $\exists!\, f \in C^\infty(N)$. s.t. $g = f\circ F$

$\Rightarrow v_1(g) = v_2(g) \Rightarrow v_1 = v_2$ $g = f\circ F$

Surjective: $\forall u \in T_{F(p)}$, $\forall f \in C^\infty(N)$. $\exists! g \in C^\infty(M)$. s.t.

$\Rightarrow$ We can define $v: C^\infty(M) \to \mathbb{R}$, $g \mapsto a(f)$. It's easy to check that $v \in T_pM$ and $u = dF_p(v)$

That $dF_p$ is an isomorphism & $d(F^{-1})_{F(p)} \circ dF_p = d(F^{-1} \circ F)_p : T_pM \to T_pM$
$= d(Id_M)_p = Id_{T_pM}$

$\Rightarrow (dF_p)^{-1} = d(F^{-1})_{F(p)}$ $\square$

Prop III.1.8. Let $M$ be a smooth manifold (with boundary). $p \in M$, $v \in T_pM$. If $f, g \in C^\infty(M)$ agree on some neighbourhood of $p$, then $v(f) = v(g)$

pf: Let $h = f - g$ and let $U$ be neighbourhood of $p$ where $h$ vanishes
By partitions of unity $\exists \psi$ s.t. $\psi \equiv 1$ on $supp(h)$ and be supported in $M \setminus \{p\}$
$\Rightarrow h = \psi h$, $h(p) = \psi(p) = 0 \Rightarrow v(h) = v(\psi h) = 0$ $\square$

Prop. III.1.9. Let $M$ be a smooth manifold, $U \subseteq M$ be the open submanifold, and $\iota: U \hookrightarrow M$ be inclusion map
For every $p \in U$, show that $d\iota_p : T_pU \to T_pM$ is an isomorphism

pf: Injective: Suppose $v \in T_pU$ and $d\iota_p(v) = 0 \in T_pM$.
Let $B$ be a neighbourhood of $p$ s.t. $\overline{B} \subseteq U$
By Extension lemma of smooth functions, $\forall f \in C^\infty(U)$ $\exists \tilde{f} \in C^\infty(M)$
s.t. $\tilde{f}|_{\overline{B}} = f|_{\overline{B}}$, $supp(\tilde{f}) = U \Rightarrow \tilde{f}|_U$ and $f \in C^\infty(U)$
By III.1.8. $v(f) = v(\tilde{f}|_U) = v(\tilde{f} \circ \iota) = d\iota_p(v)(\tilde{f}) = 0 \Rightarrow v = 0$

Surjective: $\forall w \in T_pM$, define $v: C^\infty(U) \to \mathbb{R}$ satisfying $v(f) = w(\tilde{f})$. $f \equiv \tilde{f}$ on $\overline{B} \subseteq U$
Since $v(f)$ is independent of the choice of $\tilde{f}$.
It's easy to check that $v \in T_pU$. and $v(f) = v(\tilde{f} \circ \iota) = d\iota_p(v)(\tilde{f}) = w(\tilde{f})$
$\Rightarrow d\iota_p(v) = w$. $\Rightarrow d\iota_p$ is surjective $\square$.

Prop. III.1.10. (Dim of $T_pM$) If $M$ is $n$-dim smooth manifold, then $\forall p \in M$, $T_pM$ is an $n$-dim vect space.

pf: Given $p \in M$, $(U, \varphi)$ be a smooth coordinate chart containing $p$.
Then we have: $T_pM \cong T_pU \cong T_{\varphi(p)}\mathbb{R}^n \Rightarrow \dim T_pM = n$. $\square$.

Prop. III.1.11. (Dim of $T_pM$, $M$ with boundary). If $M$ is $n$-dim smooth, with boundary, then $\forall p \in M$, $T_pM$ is an $n$-dim vector space.

pf: Lemma: $\iota: \mathbb{H}^n \hookrightarrow \mathbb{R}^n$ denote the inclusion map. $\forall a \in \partial\mathbb{H}^n$, $d\iota_a: T_a\mathbb{H}^n \to T_a\mathbb{R}^n$ is an homeomorphism
① $d\iota_a(v) = 0$, $v \in T_a\mathbb{H}^n$, $f: \mathbb{H}^n \to \mathbb{R}$, Extend $f$ to $\tilde{f}: \mathbb{R}^n \to \mathbb{R}$
Then $\tilde{f} \circ \iota = f$. $\Rightarrow vf = v(\tilde{f} \circ \iota) = d\iota_a(v)(\tilde{f}) = 0 \Rightarrow v = 0$.
② Surjective, its similar. $\square$

Prop. III.1.12. Suppose $V$ is an $n$-dim real vector space. with its standard smooth structure.
$\forall a \in V$, the map $v \mapsto D_v|_a$ where $D_v|_a : C^\infty(V) \to \mathbb{R}$.
$f \mapsto \frac{d}{dt}\big|_{t=0} f(a+tv)$.

is a canonical isomorphism from $V$ to $T_aV$, s.t. $\forall$ linear map $L: V \to W$

$$\begin{array}{ccc} V & \xrightarrow{\cong} & T_aV \\ L \downarrow & & \downarrow dL_a \\ W & \xrightarrow{\cong} & T_{L(a)}W \end{array}$$

Pf: ① $v \mapsto D_v|_a$ is an isomorphism, where the proof is similar to the case of $\mathbb{R}^n$

② $dl_a(D_v|_a)(f) = D_v|_a(f\circ l_a) = \frac{d}{dt}\big|_{t=0} f(l(a+tv)) = \frac{d}{dt}\big|_{t=0} f(la+tlv) = D_{lv}|_{la}(f)$ □.

Prop Ⅱ.1.13. Let $M_1, \dots, M_k$ be smooth manifolds, and $\forall j$, $\pi_j : M_1 \times \dots \times M_k \to M_j$ be projection.

$\forall p \in M_1 \times \dots \times M_k$, the map $\alpha : T_p(M_1 \times \dots \times M_k) \longrightarrow T_{p_1}M_1 \oplus \dots \oplus T_{p_k}M_k$

defined by $\alpha(v) = (d(\pi_1)_{p_1}(v), \dots, d(\pi_k)_{p_k}(v))$ is an isomorphism.

Pf: Since $d(\pi_k)_p(v_0) = 0 \Rightarrow \forall f \in C^\infty(\pi_k)$, $v_1(f\circ\pi_k) = 0$

$\Rightarrow$

Next we'll compute the tangent vector and the differential in a concrete way.

Def Ⅲ.1.14. Let $M$ be n-dim smooth, $p \in M$ and $(U, \varphi)$ be the chart containing $p$

Then define $\frac{\partial}{\partial x^i}\big|_p := (d\varphi_p^{-1})\left(\frac{\partial}{\partial x^i}\big|_{\varphi(p)}\right) = d(\varphi^{-1})_{\varphi(p)}\left(\frac{\partial}{\partial x^i}\big|_{\varphi(p)}\right) : C^\infty(U) \to \mathbb{R}$

Rmk: ① Since $d\varphi_p : T_pU \to T_{\varphi(p)}\mathbb{R}^n$ is an isomorphism, we have $\frac{\partial}{\partial x^i}\big|_p$ forms a basis for $T_pM$ ($T_pM \cong T_pU \cong T_{\varphi(p)}\mathbb{R}^n$).

② $\frac{\partial}{\partial x^i}\big|_p(f) = \frac{\partial}{\partial x^i}\big|_{\varphi(p)}(f\circ\varphi^{-1}) := \frac{\partial \hat{f}}{\partial x^i}(\hat{p})$. $\hat{f} = f\circ\varphi^{-1}$, $\hat{p} = \varphi(p) \in \mathbb{R}^n$, $\forall f \in C^\infty(U)$.

③ In the subject, we tend to use $v = v^i \frac{\partial}{\partial x^i}\big|_p$ to represent $\sum_{i=1}^n \frac{\partial}{\partial x^i}\big|_p \cdot v^i$

④ Let $x^j : U \to \mathbb{R}$, $p \to \pi_j \circ \varphi(p)$, then $v(x^j) = \left(v^i \frac{\partial}{\partial x^i}\big|_p\right)(x^j) = v^i \frac{\partial x^j}{\partial x^i}(p) = v^j$.

Prop Ⅲ.1.15.

⑤ If $F : M \to N$ smooth, $\forall p \in M$, $(U, \varphi)$ containing $p$ and $(V, \psi)$ containing $F(p)$,

$\hat{F} := \psi \circ F \circ \varphi^{-1} : \mathbb{R}^n \to \mathbb{R}^m$. then $F\circ\varphi^{-1} = \psi^{-1}\circ\hat{F}$

Then $dF_p\left(\frac{\partial}{\partial x^i}\big|_p\right) = dF_p\left(d(\varphi^{-1})_{\hat{p}}\left(\frac{\partial}{\partial x^i}\big|_{\hat{p}}\right)\right) = d(\psi^{-1})_{\hat{F}(\hat{p})}\left(d\hat{F}_{\hat{p}}\left(\frac{\partial}{\partial x^i}\big|_{\hat{p}}\right)\right)$

$= d(\psi^{-1})_{\hat{F}(\hat{p})}\left(\frac{\partial \hat{F}^j}{\partial x^i}(\hat{p})\frac{\partial}{\partial y^j}\big|_{\hat{F}(\hat{p})}\right)$ (By rule of chain)

$= \frac{\partial \hat{F}^j}{\partial x^i}(\hat{p})\frac{\partial}{\partial y^j}\big|_{F(p)}$

## 2. The Tangent Bundle

**Def Ⅱ 2.1** $M$ be a smooth manifold, define the tangent bundle of $M$, denoted by $TM$

$$TM := \coprod_{p\in M} T_pM$$ to be the disjoint union of tangent spaces

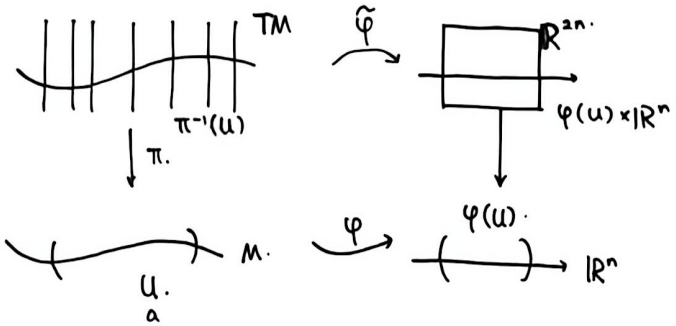
**Rmk.** ① An element of $TM$ is an ordered pair $(p, v)$. where $p \in M$, $v \in T_pM$.

② $\pi : TM \to M \quad (p, v) \mapsto p$.

The most important observation is that $TM$ can be viewed as a manifold and the projection map $\pi : TM \to M$ is smooth.

**Prop. Ⅲ.2.2.** $M$ is an $n$-dim manifold, then $TM$ is a $2n-$dim manifold and $\pi : TM \to M$ is smooth

**Pf:** Hint.




① $\forall p \in M$. and smooth chart $(U, \varphi)$ containing $p$, define the topology on $TM := \{\pi^{-1}(U) \mid U \subseteq M$ (It's easy to check $T_2$ & $A_2$)

② Denote $\varphi$ as the coordinate functions $(x^1, \dots, x^n)$.

Then define $\tilde{\varphi} : TM \to \mathbb{R}^{2n}$. $v^i \frac{\partial}{\partial x^i}\Big|_p \mapsto (x^1(p), \dots, x^n(p), v^1, \dots, v^n) = (\pi(p), v^1, \dots, v^n)$

Therefore $\tilde{\varphi}$ is a homeomorphism (since $T_pM \cong \mathbb{R}^n$). (Then $TM$ is l.c. Euclidean)

③ $\forall\ \pi^{-1}(U) \cap \pi^{-1}(V) \neq \phi$, we have $\tilde{\psi} \circ \tilde{\varphi}^{-1}(x^1, \dots, x^n, v^1, \dots, v^n) = (\tilde{x}^1, \dots, \tilde{x}^n, \frac{\partial \tilde{x}^1}{\partial x^j} v^j, \dots, \frac{\partial \tilde{x}^1}{\partial x}$

is smooth. □.

**Rmk.** $(x^i, v^i)$ is called the natural and coordinate on $TM$.

**Prop. Ⅲ.2.3.** If $M$ is an $n$-dim manifold, $M \subseteq (U, \varphi)$, then $TM \cong M \times \mathbb{R}^n$.

**Pf:** trivial □.

**Def Ⅲ.2.4.** Suppose there is a smooth map $F : M \to N$, then define $dF : TM \to TN$ as $dF|_{T_pM} := dF_p$.

**Prop.** $dF : TM \to TN$ is smooth

**Pf:** $dF : TM \to TN \quad (x^1, \dots, x^n, v^1, \dots, v^n) \mapsto (F^1(x), \dots, F^n(x), \dots, \frac{\partial F^1}{\partial x^i}(x) v^i, \dots, \frac{\partial F^n}{\partial x^i}(x) v^i)$

is smooth □.

Actually, there is a notation trick.

$$\begin{array}{ccc} TM & \xrightarrow{dF} & TN \\ \tilde{\varphi}\downarrow & & \downarrow\tilde{\Phi} \\ \mathbb{R}^n\times\mathbb{R}^n & \longrightarrow & \mathbb{R}^n\times\mathbb{R}^n \end{array}$$

used similarly as

$\tilde{\Phi} \circ dF \circ \tilde{\varphi}^{-1}$.

: $dF_p : T_pM \to T_pN$ has the formula.

: $dF_p\left(\frac{\partial}{\partial x^i}\Big|_p\right) = \frac{\partial \hat{F}^j}{\partial x^i}(\hat{p}) \frac{\partial}{\partial y^j}\Big|_{F(p)}$ where $\begin{cases} \hat{F} := \Psi \circ F \circ \varphi^{-1} : \mathbb{R}^n \to \mathbb{R}^n. \\ (y^1, \dots, y^n) \text{ be the coordinate function} \end{cases}$

Prop III.1.15 $M$ be smooth. $(U,\varphi)$ & $(V,\psi)$ are smooth charts. $p \in U \cap V$.

And the coordinate functions of $\varphi$ by $(x^i)$, $\psi$ by $(\tilde{x}^i)$

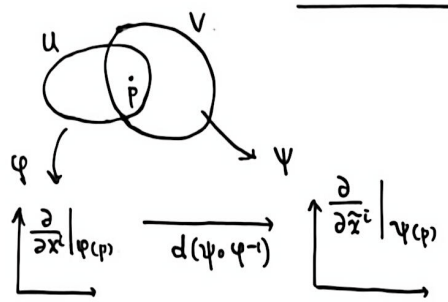
Then the transition map $\psi \circ \varphi^{-1}: \varphi(U\cap V) \to \psi(U\cap V)$ $\quad x \mapsto (\tilde{x}^1(x) \cdots \tilde{x}^n(x))$

Then $d(\psi\circ\varphi^{-1})_{\varphi(p)}\left(\frac{\partial}{\partial x^i}\Big|_{\varphi(p)}\right) = \frac{\partial \tilde{x}^j}{\partial x^i}(\varphi(p))\,\frac{\partial}{\partial \tilde{x}^j}\Big|_{\psi(p)}$

Therefore. $\underline{\frac{\partial}{\partial x^i}\Big|_p} = d(\varphi^{-1})_{\hat{p}}\left(\frac{\partial}{\partial x^i}\Big|_{\hat{p}}\right) = d(\psi^{-1})_{\psi(p)} \circ d(\psi\circ\varphi^{-1})_{\hat{p}}\left(\frac{\partial}{\partial x^i}\Big|_{\hat{p}}\right)$

$$= d(\psi^{-1})_{\psi(p)}\left(\frac{\partial \tilde{x}^j}{\partial x^i}(\varphi(p))\frac{\partial}{\partial \tilde{x}^j}\Big|_{\psi(p)}\right)$$

$$= \underline{\frac{\partial \tilde{x}^j}{\partial x^i}(\hat{p})\,\frac{\partial}{\partial \tilde{x}^j}\Big|_p} \quad (*)$$




Apply $(*)$ to $v = v^i \frac{\partial}{\partial x^i}\Big|_p \Rightarrow$ we have $\underline{\tilde{v}^j = \frac{\partial \tilde{x}^j}{\partial x^i}(\hat{p})v^i.}$ $(**)$

$= \tilde{v}^j \frac{\partial}{\partial \tilde{x}^j}\Big|_p.$

Example: $M = \mathbb{R}^2$. $(x,y) = (r\cos\theta, r\sin\theta)$ Let $v \in T_p\mathbb{R}^2$ be the tangent vector whose representation is $v = 3\frac{\partial}{\partial r}\Big|_p - 2\frac{\partial}{\partial \theta}\Big|_p$. where $p = (2, \frac{\pi}{2})$

By $(**)$, $\frac{\partial}{\partial r}\Big|_p = \cos(\frac{\pi}{2})\frac{\partial}{\partial x}\Big|_p + \sin(\frac{\pi}{2})\frac{\partial}{\partial y}\Big|_p = \frac{\partial}{\partial y}\Big|_p$

$\frac{\partial}{\partial \theta}\Big|_p = -2\frac{\partial}{\partial x}\Big|_p$

$\Rightarrow v = 3\frac{\partial}{\partial y}\Big|_p + 2\frac{\partial}{\partial x}\Big|_p$ □

The next perspection of viewing tangent vectors is: the velocity of some smooth curve on $M$

Def Ⅲ.2.5. ① A smooth curve $\gamma$ on $M$ is a smooth map from $J \subseteq \mathbb{R} \to M$. ($J$ is an Interval)

② The velocity of $\gamma$ at $t_0$ ($t_0 \in J$) is the vector $\gamma'(t_0) := d\gamma\left(\frac{d}{dt}\big|_{t_0}\right) \in T_{\gamma(t_0)}M$.

Rmk. ① From the definition, we have $\gamma'(t_0)f = d\gamma\left(\frac{d}{dt}\big|_{t_0}\right)f = \frac{d}{dt_0}\big|_{t_0}(f\circ\gamma) = (f\circ\gamma)'(t_0)$ $\forall f \in C^\infty M$.

② Let $(U,\varphi)$ be smooth charts with coordinate function $(x^i)$. write $\gamma(t) = (\gamma^1(t),\dots,\gamma^n(t))$

then we have $\gamma'(t_0) = \frac{d\gamma^i}{dt}(t_0)\,\frac{\partial}{\partial x^i}\big|_{\gamma(t_0)}$.

Prop. Ⅲ.2.6. $M$ is an $n$-dim manifold, and $p \in M$. Every $v \in T_pM$ is the velocity of some smooth curve on $M$.

pf: $\forall p \in M$, let $(U,\varphi)$ centered at $p$, with $v \in T_pM = v^i\frac{\partial}{\partial x^i}\big|_p$.

Let $\gamma: (-\varepsilon,\varepsilon) \to U$. with the coordinate representation $\gamma(t) = (tv^1,\dots,tv^n)$.

Then $\gamma(0) = \varphi^{-1}(0) = p$. , $\gamma'(0) = v^i\frac{\partial}{\partial x^i}\big|_p = v$. $\Rightarrow$ $v$ is the velocity of $\gamma$ □.

Prop. Ⅲ.2.7. Let $F: M \to N$ smooth and let $\gamma: J \to M$ be smooth curve. $\forall t_0 \in J$, we have

$F\circ\gamma: J \to N$ smooth, and $(F\circ\gamma)'(t_0) = dF(\gamma'(t_0))$.

pf: $(F\circ\gamma)'(t) = d(F\circ\gamma)\left(\frac{d}{dt}\big|_{t_0}\right) = dF\circ d\gamma\left(\frac{d}{dt}\big|_{t_0}\right) = dF(\gamma'(t_0))$ □.

Coro. Ⅲ.2.8. $F: M \to N$ smooth, $p \in M$ and $v \in T_pM$. Then $dF_p(v) = (F\circ\gamma)'(0)$

for any smooth curve $\gamma: J \to M$ s.t. $0 \in J$, $\gamma(0) = p$, $\gamma'(0) = v$. □.

3. More alternative definitions of the Tangent space.